EREN YILDIZ, PH.D.

Postdoctoral Researcher

eyildiz8@gatech.edu 404-369-9869 Atlanta, United States
erenyildiz33.github.io erenyildiz33 erenyildiz33 0000-0002-4631-7834
scholar.google.com/citations?user=LXUvnL0AAAAJ&hl=en

I am a Postdoctoral Researcher at the **Georgia Institute of Technology**, specializing in **batteryless sensing platforms, energy-harvesting, and sustainable embedded systems**. My research advances the foundations of ultra–low-power and intermittent computing through resilient hardware/software co-designs, adaptive energy-aware runtimes, and new architectural mechanisms that enable long-lived, self-powered sensing systems. By leveraging ambient energy sources such as RF, solar, and vibration, I develop scalable and environmentally responsible IoT platforms that operate reliably under real-world, energy-variable conditions.

I have authored more than 10 peer-reviewed publications in top-tier venues such as **OSDI**, **EuroSys**, **SenSys**, and **EWSN**, and I was recently honored with the **EWSN 2025 Rising Star Award**. My work contributes to the development of energy-efficient embedded intelligence for smart infrastructure, healthcare, and environmental monitoring, and lays the foundation for the future of carbon- and energy-aware, intelligent, batteryless computing.

PERSONAL DETAILS

Nationality: Turkish
Date of Birth: 1989
Languages: Turkish (native), English (Academic Proficiency), Spanish (beginner)

EXPERTISE

Embedded Systems | Operating Systems/Runtimes | Computer Architectures | Low-power Architectures | Energy Harvesting | Intermittent Computing | Tiny Machine Learning | Sustainable IoT Systems

PUBLICATION RECORD

Selected Publications

MobiSys 2026: *BIONIC: Power System Architecture for Time-Sensitive Intermittent Computing.* ***Eren Yıldız*** et al. The ACM International Conference on Mobile Systems, Applications, and Services.

MobiSys 2026: *MicroGreen: A Framework on Carbon-aware Edge ML System Design.* Xuesi Chen, Ilan Mandel, ***Eren Yıldız*** et al. The ACM International Conference on Mobile Systems, Applications, and Services.

EWSN 2025: *PEARL: Power- and Energy-Aware Multicore Intermittent Computing.* Khakim Akhunov, ***Eren Yıldız*** et al. The 22nd International Conference on Embedded Wireless Systems and Networks.

TECS 2025: *CapDYN: Adaptive Self-Scaling Energy Storage for Powering Batteryless IoT.* Maria Doglioni, ***Eren Yildiz*** et al. ACM Transactions on Embedded Computing Systems.

SenSys 2024: *Fast-Inf: ultra-fast embedded intelligence on the batteryless edge.* Leonardo Lucio Custode, Pietro Farina, ***Eren Yildiz*** et al. The 22nd ACM Conference on Embedded Networked Sensor Systems.

EuroSys 2024: *Adaptable runtime monitoring for intermittent systems.* ***Eren Yıldız*** et al. Th 19th European Conference on Computer Systems.

EWSN 2024: *Memory-efficient energy-adaptive inference of pre-trained models on batteryless embedded systems.*Pietro Farina, Subrata Biswas, ***Eren Yıldız*** et al. The 21st International Conference on Embedded Wireless Systems and Networks.

EuroSys 2023: *Efficient and safe i/o operations for intermittent systems.* ***Eren Yıldız*** et al. The 18th European Conference on Computer Systems.

TECS 2022: *Etap: Energy-aware timing analysis of intermittent programs.* Ferhat Erata, ***Eren Yildiz*** et al. ACM Transactions on Embedded Computing Systems.

OSDI 2022: *Immortal threads: Multithreaded event-driven intermittent computing on Ultra-Low-Power microcontrollers.* ***Eren Yıldız*** et al. In 16th USENIX Symposium on Operating Systems Design and Implementation.

Just Submitted

Patent: *A New Power System Architecture for Batteryless Systems.* ***Eren Yıldız*** et al. Italian Patent and Trademark Office (Under Evaluation).

RESEARCH ARTIFACTS

Immortal Threads: https://github.com/tinysystems/ImmortalThreads
Immortal Threads is a novel programming model that brings pseudo-stackful multithreaded processing to intermittent computing. Programmers using Immortal Threads are oblivious to intermittent execution and write their applications in a multithreaded fashion using common event-driven multithreading primitives.

EaseIO: https://github.com/tinysystems/easeIO
EaseIO is a novel programming model that introduces re-execution semantics of IO operations for intermittent computing. Re-executed IO operations in intermittent systems' tasks might lead to memory inconsistencies and high energy consumption. EaseIO offers new interfaces to call IO functions to overcome memory inconsistencies and save energy.

ARTEMIS: https://github.com/tinysystems/ARTEMIS
ARTEMIS is the first framework designed to facilitate flexible property checking of intermittent programs at runtime. In ARTEMIS, developers use a language to define desired properties of the intermittent system, focusing on system properties rather than implementation specifics.

PEARL: https://github.com/tinysystems/PEARL
PEARL is a novel systems support that can make existing multicore microcontroller (MCU) platforms suitable for efficient intermittent computing.

FreeML: https://github.com/tinysystems/FreeML
FreeML is a framework to optimize pre-trained DNN models for memory-efficient and energy- adaptive inference on batteryless systems.

SERVICE RECORD

Journal Reviews

- ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- IEEE Internet of Things Journal
- Future Generation Computer Systems
- IEEE Transactions on Very Large Scale Integration Systems
- IEEE Transactions on Sustainable Computing

Conference/Workshop TPCs

- 14th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (ENSsys'26)

As an External Reviewer

- The ACM SIGBED International Conference on Embedded Software (EMSOFT)

ACHIEVEMENTS & AWARDS

EWSN Rising Star 2025.
I selected as one of the Rising Star in Conference on Embedded Wireless Systems and Networks (EWSN) 2025.

Research Fellowship Award
2214-A - International Research Fellowship Programme for PhD Students. The Scientific And Technological Research Council Of Turkey(TUBITAK), 20,000$.

EMPLOYMENT AND APPOINTMENTS

Postdoctoral Researcher **Feb. 2025 – Ongoing**
Georgia Institute of Technology, College of Computing, School of Interactive Computing, Atlanta, United States

Research Assistant Dr. **Oct. 2023 – Feb. 2025**
RTEU, Department of Computer Engineering, Rize, Turkiye

Visiting Researcher **Jul. 2024 – Aug. 2024**
University of Trento, Department of Information Engineering and Computer Science, Trento, Italy

Visiting Researcher **Dec. 2021 – Dec. 2022**
Northwestern University, Computer Science Department, Chicago, United States

Research Assistant **Oct. 2018 – Oct. 2023**
Ege University, Department of Computer Engineering, Izmir, Turkiye

R&D Engineer **Sept. 2015 – Feb. 2017**
Emtech Information Technologies, Mersin, Turkiye

EDUCATION

Ph.D. in Computer Engineering

Ege University

2018 – 2023

Thesis title: System Support for Efficient Operation of Batteryless Sensor Systems

M.Sc. in Electrical Electronics Engineering

Mersin University

2015 – 2018

Thesis title: Cyberbullying Detection And Classification With Artificial Intelligence Algorithms In Social Network

B.Sc. in Electrical Electronics Engineering

Mersin University

2009 – 2014

COURSES TAUGHT

Courses at the Department of Computer Engineering, RTEU, Turkiye

CE-202 Signals and Systems
2024 - Spring Semester

CE-414 Tiny Machine Learning
2024 - Spring Semester

CEN-109 Fundamental Information Technologies
2024 - Fall Semester

TEACHING ASSISTANT EXPERIENCE

Courses at the Department of Computer Engineering, Ege University, Turkiye

Digital Computer Design
2019 - 2023 Spring Semester

Logic Design
2019 - 2023 Fall Semester

Digital Electronics
2019 - 2023 Spring Semester

Electrical Circuits
2019 - 2023 Fall Semester

Microprocessors
2019 - 2023 Fall Semester

Microcontroller-Based System Design-
2019 - 2023 Spring Semester

MENTORING

Undergraduate Students Mentoring

I have supervised several undergraduate students whose work has resulted in high-impact outcomes.

- **Davide Cavedon & Stefano Putelli (University of Trento):** Co-developed a new energy storage and management architecture for batteryless systems, leading to an **MobiSys'26 submission and a patent submission**.
- **Pietro Farina (University of Trento):** Worked on TinyML optimization for intermittent systems, **resulting in a publication at EWSN'24**.
- **Lijun Chen (University of Trento):** Worked on compiler front-end design of Immortal Threads runtime, **resulting in a publication at OSDI'22**.

PhD Students Mentoring

- **Summit Shrestha (Georgia Tech):** Co-mentoring research on energy- and carbon-efficient LLM execution on edge and intermittent devices.
- **Nikola Maruszewski (Georgia Tech):** Co-mentoring research on MTJ-based spintronic elements for sustainable, ultra-low-power IoT architectures.

This demonstrates my ability to guide students at different levels toward meaningful contributions, including top-tier submissions, patents, and interdisciplinary research directions.

TEACHING INTERESTS

Computer Architecture | Operating Systems | Embedded Systems | Digital Logic Design | Digital Systems | Microprocessors & Microcontrollers | Internet-of-Things Sensors, Systems, and Applications | Wearable Systems | Real-Time Operating Systems | Tiny Machine Learning for Embedded Systems

PROFESSIONAL MEMBERSHIPS

HiPEAC (High Performance Embedded Architecture and Compilation)
https://www.hipeac.net

Publications

Journal Articles

- X. Chen, A. Goldner, **E. Yildiz**, *et al.*, "From component to system: Rethinking edge computing design through a carbon-aware lens," ACM SIGENERGY Energy Informatics Review, vol. 5, no. 2, pp. 125–131, 2025.
- M. Doglioni, **E. Yildiz**, M. Nardello, K. Akhunov, K. S. Yildirim, and D. Brunelli, "Capdyn: Adaptive self-scaling energy storage for powering batteryless iot," ACM Transactions on Embedded Computing Systems (TECS), 2024.
- F. Erata, **E. Yildiz**, A. Goknil, *et al.*, "Etap: Energy-aware timing analysis of intermittent programs," ACM Transactions on Embedded Computing Systems (TECS), 2022.
- A. Cigdem, **E. Yildiz**, and E. S. Essiz, "Automatic detection of cyberbullying in formspring. me, myspace and youtube social networks," Turkish Journal of Engineering, vol. 3, no. 4, pp. 168–178, 2019.

Conference Proceedings

- K. Akhunov, **E. Yildiz**, and K. S. Yıldırım, "Pearl: Power- and energy-aware multicore intermittent computing," in Proceedings of the 2025 International Conference on Embedded Wireless Systems and Networks (EWSN'25).
- P. Farina, S. Biswas, **E. Yildiz**, *et al.*, "Memory-efficient energy-adaptive inference of pre-trained models on battery-less embedded systems," in Proceedings of the 2024 International Conference on Embedded Wireless Systems and Networks (EWSN'24).
- S. Ahmed, **E. Yildiz**, S. Holla, *et al.*, "Bootstrapping health wearables powered by intra-body power transfer," in 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), IEEE, 2024, pp. 1–4.
- L. L. Custode, P. Farina, **E. Yildiz**, R. B. Kilic, K. S. Yildirim, and G. Iacca, "Fast-inf: Ultra-fast embedded intelligence on the batteryless edge," in Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems (SenSys'24), 2024, pp. 239–252.
- **E. Yildiz**, K. Akhunov, L. A. Riva, A. Goknil, I. Kurtev, and K. S. Yildirim, "Adaptable runtime monitoring for intermittent systems," in Proceedings of the Nineteenth European Conference on Computer Systems (EuroSys'24), 2024, pp. 1175–1191.
- **E. Yildiz**, D. Cavedon, and K. S. Yildirim, "On tracking time with better resolution and range in batteryless systems," in Proceedings of the 12th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems (EnsSys'24), 2024, pp. 42–47.
- K. Akhunov, **E. Yildiz**, and K. S. Yildirim, "Enabling efficient intermittent computing on brand new microcontrollers via tracking programmable voltage thresholds," in Proceedings of the 11th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (EnsSys'21), 2023, pp. 16–22.
- **E. Yildiz**, S. Ahmed, B. Islam, J. Hester, and K. S. Yıldırım, "Efficient and safe i/o operations for intermittent systems," in Proceedings of the Eighteenth European Conference on Computer Systems (EuroSys 23), 2023.
- **E. Yildiz**, L. Chen, and K. S. Yıldırım, "Immortal threads: Multithreaded event-driven intermittent computing on {ultra-low-power} microcontrollers," in 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22), 2022, pp. 339–355.
- **E. Yildiz** and K. S. Yıldırım, "Persistent timekeeping using harvested power measurements," in Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems(SenSys'21), 2021, pp. 572–574.
- **E. Yildiz** and K. S. Yildirim, "Defragmenting energy storage in batteryless sensing devices," in Proceedings of the 8th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems(EnsSys'20), 2020, pp. 36–42.
- **E. Yildiz**, K. S. Yıldırım, P. Pawelczak, and J. Hester, "On the accuracy of network synchronization using persistent hourglass clocks," in Proceedings of the 7th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (EnsSys'19), 2019, pp. 35–41.
- **E. Yildiz**, C. Aci, and E. S. Essiz, "The effects of attribute selection in artificial neural network based classifiers on cyber-bullying detection," in 2018 3rd International Conference on Computer Science and Engineering (UBMK), IEEE, 2018, pp. 6–11.
- **E. Yildiz**, Ç. Acı, and E. S. Eşsiz, "Performance analysis of artificial neural network based classfiers for cyberbulling detection," in 2018 3rd International Conference on Computer Science and Engineering (UBMK), IEEE, 2018, pp. 1–5.

References

Assoc. Prof. Josiah Hester
Georgia Institute of Technology, United States
josiah@gatech.edu
https://josiahhester.com

Assoc. Prof. Kasim Sinan Yildirim
University of Trento, Italy
kasimsinan.yildirim@unitn.it
https://sinanyil81.github.io

Assoc. Prof. Ivan Kurtev
Eindhoven University of Technology, Netherlands
i.kurtev@tue.nl
https://www.tue.nl/en/research/researchers/ivan-kurtev

Assist. Prof. Udit Gupta
Cornell Tech, Cornell University, United States
ugupta@cornell.edu
https://ugupta.com